JPE Replication Report

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1 Paper Identification

ID	20180648
Author	Tincani
Title	Peer Effects and Rank Concerns in the Classroom
Iteration	1

Some useful information up front:

- JPE Data and Code Availability Policy
- Step by step guidance from Data Editor for Authors
- Template README

2 Summary

In assessing compliance with our Data and Code Availability Policy, our reproducibility team has identified the following issues, which we ask you to address:

1. There are a few datasets which are not cited in the readme and the paper.
2. Regarding the confidential data, I am proposing a few things which we normally add to the readme - please let me know what is feasible here for you.
3. We found a few minor bugs detailed below, please check those.

3 General

- [x] Data Availability and Provenance Statements
  - [x] Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
  - [x] Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
  - [ ] License for Data (optional, but recommended)
  - [x] Details on each Data Source
- [x] Dataset list
- [x] Computational requirements
  - [x] Software Requirements
  - [x] Controlled Randomness (as necessary)
  - [x] Memory, Runtime, and Storage Requirements
- [x] Description of programs/code
  - [x] License for Code (Optional, but recommended)
  - [x] Details (as necessary)
- [x] List of tables and programs
- [ ] References (Optional, but recommended)

Note: On controlled randomness. The master do-file sets set seeds for replicability.

4 High-level Description of Research Project

This research project...

- [x] uses observational data.
- [x] uses *confidential* observational data.
- [ ] uses data generated via experiment or trial by the authors (in field or lab).
  - [ ] For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
  - [ ] A PDF copy of IRB approval is contained in the package.
  - [ ] A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
  - [ ] A PDF copy of the instructions given to participants in both original language and an English translation is included.
- [ ] uses data generated via survey by the authors (in field or online)
  - [ ] The programs used to administer the survey (e.g. qualtrics survey file) are included.
  - [ ] A PDF copy of IRB approval is contained in the package.

- [ ] A PDF of the original questionaire(s) and information on the selection and eligibility of participants is included.
- [ ] is a simulation study: the data are generated by an algorithm.
- [ ] is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

Note: The data uses survey data, but it is not generated by the authors.

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

> {REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

> {REQUIRED} Please provide affirmation of support for replication checks. - As per the JPE policy, "In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication."

> {REQUIRED} Please provide a codebook for the data. - The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

Replicator's notes:

- **Preservation duration:** The README does **not** specify how long the restricted-access data will be preserved.
- **Affirmation of support for replication checks:** Section 3.1 of the README affirms the author has "legitimate access to and permission to use all data employed in this manuscript," but does not explicitly commit to the JPE 5-year preservation + support obligation.
- **Codebook:** The README provides a Data Inventory (Section 3.2) describing each dataset at a general level (e.g., "standardized test score data for fourth grades in 2005 and 2007"), but does not provide a variable-level codebook for any of the restricted datasets.

5.1 Data Sources

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
Rendimiento por Estudiante (Ministerio de Educación; 6 CSVs, 2005–2011)	no*	yes	yes	yes	yes	yes
Matrícula por Estudiante 2009 (Ministerio de Educación)	yes	no	yes	yes	yes	no
CPV2002 Census: PERSONA + Vivienda (INE Chile)	yes	no	yes	yes	yes	yes
SIMCE Puntajes por Estudiante (Agencia de Calidad; 4 files)	no	yes	yes	yes	yes	yes
SIMCE Cuestionarios Estudiantes (Agencia; 2 files)	no	yes	yes	yes	yes	yes
SIMCE Cuestionarios Apoderados (Agencia; 4 files)	no	yes	yes	yes	yes	yes
SIMCE Cuestionarios Docentes (Agencia; 4 files)	no	yes	yes	yes	yes	yes
SEP School Expenditure Reports — SS_Rendidos (Ministerio)	no	yes	no	yes	yes	yes
GSHHG shoreline (NOAA; .shp + .shx)	yes	no	yes	yes	yes	no

file/name	public	private	DOI/URL	Access described	cited readme	cited paper
USGS earthquake catalog (`earthquake_catalog_6.csv`)	yes	no	yes	yes	yes	yes
Natural Earth 110m Admin-0 Countries	yes	no	yes	yes	yes	no
Schools closed/receiving — `EE_cerr_201011.csv` (author)	yes	no	N/A	yes	yes	yes
Geocoded municipality coords — `comuna_alu_geocoded.csv` (author)	yes	no	N/A	yes	yes	no
School/student-to-asperity distances — `sch_com_dist.csv`, `stud_com_dist.csv` (author)	yes	no	N/A	yes	yes	yes
Aggregated expenditure categories — `expcat_consulting.csv`, `expcat_other.csv` (author)	yes	no	N/A	yes	yes	no

Note: The Rendimiento files are described as "publicly available" in the README (Note 1) but are placed in `confidential-data-not-for-publication/raw/`, not in `data/raw/`. See breakdown below.

5.1.1 Rendimiento por Estudiante (Ministerio de Educación, Chile)

Files: `20130425_Rendimiento_2005_20060316_PUBL.csv`, `20130226_Rendimiento_2007_20080312_PUBL.csv`, `Rendimiento por estudiante 2008.csv`, `20130227_Rendimiento_2009_20100715_PUBL.csv`, `Rendimiento por estudiante 2010.csv`, `20130301_Rendimiento_2011_20120416_PUBL.csv`. All located in `confidential-data-not-for-publication/raw/`.

- [ ] Dataset is provided in public deposit
- [x] Dataset is PRIVATELY provided (not for publication)
- [x] DOI or URL is provided, and works [NOT click-verified]: https://datosabiertos.mineduc.cl/rendimiento-por-estudiante-2/
- [x] Access conditions are described:
  - Per README Note (1): dataset is publicly available on the Ministry portal, but the analysis uses specific snapshots (2005/2007/2009/2011 accessed 2013; 2010 accessed 2018; 2008 accessed 2021), and subsequent portal updates may differ.
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
  - [x] in the manuscript (referenced as "enrollment and grade registries (Rendimiento)" in the data section)
  - [x] in the README:

> Ministerio de Educación, Chile. n.d. *Rendimiento por Estudiante, 2005–2011* [dataset]. Datos Abiertos MINEDUC. Years 2005, 2007, 2009, 2011 downloaded 2013; year 2010 downloaded 2018; year 2008 downloaded 2021. https://datosabiertos.mineduc.cl/rendimiento-por-estudiante-2/

Note: The README describes this dataset as "publicly available" but ships it only in the confidential (privately-shared) folder rather than in `data/raw/`.

5.1.2 SEP School Expenditure Reports (SS_Rendidos)

Files: `SS_Rendidos_2008-2010_20120704_redacted.csv` and `SS_Rendidos_2008-2010_20120704_redacted.dta` in `confidential-data-not-for-publication/raw/`.

- [ ] Dataset is provided in public deposit
- [x] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided — no URL or DOI is given; the README only states the dataset is a restricted administrative dataset provided by Ministerio de Educación.
- [x] Access conditions are described:
  - Restricted; access subject to Ministry approval. Per README Note (5), the version shared with the JPE Data Editor is redacted of one variable (written descriptions of single expenditure items) as per an approved exemption.
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
  - [x] in the manuscript (school spending / SEP funds referenced in the results and mechanisms sections)
  - [x] in the README:

> Ministerio de Educación, Chile. n.d. *SEP School Expenditure Reports (SS_Rendidos_2008–2010_20120704)* [dataset]. Ministerio de Educación. Restricted administrative data; accessed 2012; access subject to Ministry approval.

5.1.3 CPV2002 Census (INE Chile)

Files: `CPV2002_PERSONA.csv`, `CPV2002_Vivienda.csv` in `data/raw/`.

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [x] DOI or URL is provided, and works [NOT click-verified]: https://www.ine.gob.cl/
- [x] Access conditions are described:
  - "Public microdata; official access via INE application process." Author downloaded in 2020 and 2021.
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
  - [x] in the manuscript (referenced as "2002 census" from the Chilean National Statistics Institute)
  - [x] in the README:

> Instituto Nacional de Estadísticas (INE), Chile. 2002. *Censo de Población y Vivienda 2002: Microdatos (Personas y Viviendas)* [dataset]. Instituto Nacional de Estadísticas. Access via official INE application process; accessed 2020 and 2021. https://www.ine.gob.cl/

5.1.4 SIMCE datasets (Puntajes, Cuestionarios Estudiantes/Apoderados/Docentes)

All SIMCE datasets share the same access URL: https://informacionestadistica.agenciaeducacion.cl/#/bases

- [ ] Dataset is provided in public deposit
- [x] Dataset is PRIVATELY provided (not for publication)
- [x] DOI or URL is provided, and works [NOT click-verified]: https://informacionestadistica.agenciaeducacion.cl/#/bases
- [x] Access conditions are described:
  - "Restricted; access subject to Agency approval."
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
  - [x] in the manuscript (SIMCE test scores and questionnaires referenced throughout)
  - [x] in the README (four separate citation entries under Section 3.3)

5.1.5 USGS earthquake catalog

File: `data/raw/earthquake_catalog_6.csv`.

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [x] DOI or URL is provided, and works [NOT click-verified]: https://earthquake.usgs.gov/earthquakes/search/
- [x] Access conditions are described:
  - Publicly available. Query: magnitude ≥ 6, worldwide, 1900–September 2023; downloaded 9 September 2023.
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
  - [x] in the manuscript
  - [x] in the README:

> United States Geological Survey. n.d. *USGS Earthquake Catalog (Magnitude ≥ 6, Worldwide, 1900–September 2023)* [dataset]. USGS. Downloaded 9 September 2023. https://earthquake.usgs.gov/earthquakes/search/

5.1.6 Author-constructed datasets

Files: `EE_cerr_201011.csv`, `comuna_alu_geocoded.csv`, `sch_com_dist.csv`, `stud_com_dist.csv`, `expcat_consulting.csv`, `expcat_other.csv` (all in `data/raw/`).

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [ ] DOI or URL is provided — N/A (included with the package)
- [x] Access conditions are described:
  - "Included in replication package." `EE_cerr_201011.csv` is derived from restricted Ministerio de Educación records.
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
  - [~] in the manuscript — partial: `EE_cerr_201011.csv` is referenced (paper describes obtaining "the list of schools that closed") and the distance files `sch_com_dist.csv` / `stud_com_dist.csv` are cited via `\cite{tincani_distance_measures}` in `code/tex/Paper/data.tex`; `comuna_alu_geocoded.csv` and `expcat_consulting.csv` / `expcat_other.csv` are not explicitly cited.
    - [x] in the README (four author-constructed entries under Section 3.3)

5.1.7 Public geographic data (GSHHG, Natural Earth)

Files: `data/raw/GSHHS_f_L1.shp`, `data/raw/GSHHS_f_L1.shx`, `data/raw/ne_110m_admin_0_countries.zip`.

- [x] Dataset is provided in public deposit
- [ ] Dataset is PRIVATELY provided (not for publication)
- [x] DOI or URL is provided, and works [NOT click-verified]:
    - GSHHG: https://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/
    - Natural Earth: https://naciscdn.org/naturalearth/110m/cultural/ne_110m_admin_0_countries.zip
- [x] Access conditions are described:
    - Publicly available. GSHHG downloaded 28 March 2025; Natural Earth downloaded 24 February 2026.
- [ ] Data are not cited, but a working paper, article, or other document is cited
- [x] Data are cited
    - [ ] in the manuscript (not explicitly referenced — these are used only for the map in Figure 7, and the paper does not cite the shoreline/country-boundary datasets formally)
    - [x] in the README:

> National Centers for Environmental Information (NOAA). n.d. *Global Self-consistent Hierarchical High-resolution Geography (GSHHG), latest version* [dataset]. NOAA. Downloaded 28 March 2025. https://www.ngdc.noaa.gov/mgg/shorelines/data/gshhg/latest/
>
> Natural Earth. n.d. *Natural Earth 1:110m Cultural Vectors – Admin 0 Countries* [dataset]. Natural Earth. Downloaded 24 February 2026. https://naciscdn.org/naturalearth/110m/cultural/ne_110m_admin_0_countries.zip

6 Requirements

- [x] README is in TXT, MD, or PDF format
- [x] README is accessible at the root of the package (not inside a folder)
- [ ] Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- [ ] Authors (with affiliations) are listed in the same order as on the paper

> {REQUIRED} Please review the title of the deposit as per our guidelines.

> {REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

Note: The README lists the author as "Michela M. Tincani" but without affiliations. The title does not conform to JPE guidance.

6.1 Stated Requirements

- [ ] No requirements specified
- [x] Software Requirements specified as follows:
    - **Operating System:** README Section 4.1 states the package was tested on **Windows 10 only**. No testing on macOS or Linux is reported.
    - **Python 3.12** (tested 3.12.10). Required packages with versions:
        - `pandas==2.3.3`, `numpy==2.3.4`, `geopandas==1.1.1`, `pyogrio==0.11.1`, `pyproj==3.7.2`, `shapely==2.1.2`, `geopy==2.4.1`, `matplotlib==3.10.7`, `geodatasets==2024.8.0`
        - Plus transitive dependencies listed explicitly in the README.
    - **Stata**:
        - **StataNow 19.5** for all do-files except one.
        - **Stata 15** for `21_conley.do` only.
        - All do-files specify `version 18`, except `21_conley.do` (`version 15`) and `9_lca_all.do` (`version 15` for `gsem`).
    - **R 4.5.2** (2025-10-31 ucrt). Packages: `ggplot2` 4.0.2, `dplyr` 1.2.0.
- [x] Computational Requirements specified as follows:
    - **Local (main pipeline):** Intel Core Ultra 7 268V 2.20 GHz, 32 GB RAM (31.6 GB usable), 64-bit x64-based processor.
    - **Cluster (Conley only):** UCL Myriad HPC. Modules: `beta-modules`, `gcc-libs/9.2.0`, `libpng/1.6.37/gnu-9.2.0`, `stata/15`, plus `gerun`, `emacs/28.1`, `userscripts/1.5.0`, `rcps-core/1.0.0`, `mpi/intel/2018/update3/intel`. Sun Grid Engine resource requests: 188 GB memory, 300 GB scratch, 1 OpenMP thread.
    - **Memory:** Main pipeline runs with 32 GB RAM locally. Conley estimation requires ~188 GB (cluster-scale).
- [x] Time Requirements specified as follows:
    - Full Stata pipeline excluding Conley: **~7 hours** locally.
    - `21_conley.do` on UCL Myriad: **~61 hours**.
    - `calculate_coastal_proximity.py`: < 10 minutes.
    - `create_map.py`: < 1 minute.
    - `plot_histogram_earthquakes.R`: < 1 minute.
- [x] Requirements are complete.

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 14 - Variables flagged in data: 60 - Code files with PII references: 42 - PII references in code: 4586

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	`census_parents_clean_2023.dta`	5	father, mother, degree, child, school, house
Data	`census_parents_clean_wclass_all.dta`	5	father, mother, degree, child, school, house
Data	`census_with_class_type_true_predicted_all.dta`	5	father, mother, degree, child, school, house
Data	`comuna_alu_geocoded.csv`	2	lat, lon
Data	`comuna_alu_geocoded.dta`	2	lat, lon
Data	`data_final_for_regs_all.dta`	12	school, father, mother, house, loc
Data	`data_final_for_regs_teach_all.dta`	12	school, father, mother, house, loc
Data	`data_for_map_2025_all.csv`	2	lat, lon
Data	`donut_graph_construction_types.xlsx`	1	census, name
Data	`earthquake_catalog_6.csv`	3	lat, lon, loc, location
Data	`eb_map_iv_f.txt`	1	name
Data	`post_for_predict_2025.dta`	4	gender, father, mother, house
Data	`pre_for_predict_2025.dta`	4	gender, father, mother, house
Data	`towns_with_distance_to_coast.csv`	2	lat, lon
Code	`0_Peer_Effects_and_Rank_Concerns.do`	18	name, loc, school, census, minute, lat
Code	`10_predict_vulnerability.do`	21	census, son, dob, second, name
Code	`11_geog_coastal_proximity.do`	17	school, house, name, loc
Code	`12_gen_variables.do`	95	lat, dob, son, name, school, loc, lon
Code	`13_gen_data_for_map.do`	17	name, lat, lon, coord
Code	`14_analyze_census.do`	38	house, lat, child, school, census, name, loc, dob, father, lon, mother
Code	`15_descriptive_tables_figures.do`	108	school, name, house, lon, loc, lname, lat, father, mother
Code	`16_main_effects_and_rob.do`	36	school, name, loc, lat, municipality
Code	`17_het_effects.do`	38	school, name, loc, lon

File Type	File	Variables/References	PII Categories
Code	`18_med_relationships.do`	30	school, name, loc, lat
Code	`19_school_resources.do`	56	school, sex, son, lat, loc, name, lname
Code	`1_geog_codes_rendimiento.do`	2	name
Code	`20_identifying.do`	64	school, name, loc, lname
Code	`21_conley.do`	18	lat, minute, son, name, lon
Code	`2_clean_census.do`	83	house, son, census, child, school, father, mother, wife, husband, degree, name
Code	`3_clean_simce_4alu_8alu.do`	9	school, birth, name, second
Code	`5_clean_4cpad_8cpad.do`	38	father, mother, house, name, second, son
Code	`6_clean_simce_8prof.do`	12	name, school
Code	`7_clean_rendimiento.do`	16	school, name, country
Code	`8_merge.do`	5	lon, school
Code	`9_lca_all.do`	7	name, census, loc
Code	`_gcorr.ado`	16	lat, loc
Code	`_grmiss2.ado`	7	loc
Code	`analysis_earthquake_effects.tex`	10	lon, lat, loc, location, school, second, country
Code	`appendix.tex`	16	house, lat, sex, school, loc, second, lon
Code	`appendix_reconstruction_policy.tex`	7	school, lat, loc, location, second
Code	`calculate_coastal_proximity.py`	10	coord, lat, lon, zip
Code	`conclusions.tex`	2	lon, school
Code	`create_map.py`	9	coord, lat, lon, zip, name
Code	`data.tex`	8	gender, house, school, lat, country, loc, lon, second
Code	`estadd.ado`	736	loc, lon, name, lname, lat
Code	`estout.ado`	1657	loc, name, son, lname, lat, block, lon
Code	`estpost.ado`	596	lat, loc, name, lon, lname, son
Code	`eststo.ado`	100	loc, lon, name
Code	`esttab.ado`	524	loc, lon, name, second
Code	`introduction.tex`	6	child, school, house, loc, lat, location, son, gender, lon
Code	`manuscript.tex`	3	lon, url
Code	`mechanisms.tex`	13	school, lat, loc, location, son, second, lon, gender, social
Code	`model_extension.tex`	42	lon, lat
Code	`ols_spatial_HAC.ado`	86	lat, lon, city, degree, loc, location, coord, name, second,

File Type	File	Variables/References	PII Categories
			son, lname
Code	`plot_histogram_earthquakes.R`	2	name
Code	`theoretical_model_rank_concerns.tex`	8	lat, school, lon, son, city

See Appendix for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-04-07T10:19:32.024

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for *all* `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (`C:\` etc)
44	4	15	3	0

7.1.3 Duplicate Files Report

We did not find any duplicate files.

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We found the following files larger than 100MB:

Filename	Size (MB)
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/CPA8VO.xlsx	138.66
/Replication_Package_Tincani-20180648/data/raw/GSHHS_f_L1.shp	153.85
/Replication_Package_Tincani-20180648/data/raw/CPV2002_Vivienda.csv	163.89
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/census_parents_clean_2023.dta	182.68
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/census_parents_clean_wclass_all.dta	203.96
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/2009 8b bbdd cpadres.csv	251.71
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/CEST8VO.csv	263.16
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/census_with_class_type_true_predicted_all.dta	267.8
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/Rendimiento por estudiante 2010.csv	397.4
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/20130227_Rendimiento_2009_20100715_PUBL.csv	400.06
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/Rendimiento por estudiante 2008.csv	412.74

Filename	Size (MB)
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/20130226_Rendimiento_2007_20080312_PUBL.csv	414.26
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/20130425_Rendimiento_2005_20060316_PUBL.csv	423.26
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/raw/20130301_Rendimiento_2011_20120416_PUBL.csv	423.31
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/data_final_for_regs_all.dta	480.37
/Replication_Package_Tincani-20180648/confidential-data-not-for-publication/processed/data_final_for_regs_teach_all.dta	485.77
/Replication_Package_Tincani-20180648/data/raw/20140805_matricula_unica_2009_20090430_PUBL.csv	499.0
/Replication_Package_Tincani-20180648/data/raw/CPV2002_PERSONA.csv	1345.61

7.2 Code description

cloc	github.com/AlDanial/cloc v 2.02 T=3.34 s (21.0 files/s, 640793.0 lines/s)			
Language	**files**	**blank**	**comment**	**code**
CSV	22	0	0	2115155
Stata	31	3098	759	16536
TeX	12	976	26	2998
Markdown	1	325	0	550
Python	2	58	34	114
Text	1	0	0	76
R	1	12	11	34
——–	——–	——–	——–	——–
SUM:	70	4469	830	2135463

- **31 Stata do-files** (~16,500 lines of Stata code). Numbered `0_*` through `21_*`, plus 8 bundled frozen user-written ados in `code/stata/ado_frozen/` (estout suite, egenmore, rmiss2, ols_spatial_HAC).
- **2 Python scripts**: `calculate_coastal_proximity.py` (preprocessing, Step 1) and `create_map.py` (post-Stata, Step 3; produces Figure 7).
- **1 R script**: `plot_histogram_earthquakes.R` (Step 4; produces Figure 2; self-contained and independent of the Stata pipeline).
- **1 Excel workbook** for manual verification of Figure 3 (donut graph): `code/excel/donut_graph_construction_types.xlsx`.
- **12 TeX files** (paper + appendix source) — not part of the reproducibility pipeline per se, but determine which outputs feed into the final compiled paper.
- [x] The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

> NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

Note: The authors provide a program-level mapping in README §11 (Figures, Tables, In-text numbers).

7.3 Computing Environment of the Replicator

The main replication was performed on a Nuvolos cloud instance provisioned by the Data Editor. Initial package setup and report-writing were performed locally on an Apple MacBook Air (M1, 8 GB RAM, macOS Sequoia 15.5), but the full computational pipeline ran on Nuvolos because of memory-concerns.

Nuvolos instance specifications:

- Linux x86_64 (conda `(base)` environment visible from terminal)

- 64 GB RAM
- Separate containers for the Stata, R, and Python applications.

Software used:

- **StataNow 19.5 SE** (Stata/SE, not /MP — some do-files emit `set processors allowed only with Stata/MP`; execution continues via `capture noisily`)
- **Python 3.12** (conda `(base)` env) with pinned packages per README §4.2: `pandas 2.3.3`, `numpy 2.3.4`, `geopandas 1.1.1`, `geopy 2.4.1`, `matplotlib 3.10.7`, `geodatasets 2024.8.0`, plus transitive dependencies
- **R 4.x** (with `ggplot2` and `dplyr` auto-installed by the script)

8 Replication

1. Conley estimation (`code/stata/21_conley.do`) skipped. Per README §8.2, this do-file requires 188 GB RAM and ~61 hours of runtime on a Stata 15 HPC cluster (UCL Myriad). This exceeds the memory of both my local machine and the default Nuvolos instance. Table A14 was verified against the author's log file (see README §6.2) rather than re-executed. To skip this step, I commented out line 218 of `code/stata/0_Peer_Effects_and_Rank_Concerns.do`.

2. Python Step 1 (`code/python/calculate_coastal_proximity.py`). Installed pinned packages on the Nuvolos `(base)` conda environment. Script ran to completion.

3. Stata master do-file: first run blocked by path-detection bug. After `cd`'ing into the package root and invoking `do code/stata/0_Peer_Effects_and_Rank_Concerns.do`, the do-file's own root-detection logic immediately redirected the working directory away from the package root (to `/home/datahub`), causing the script to exit with `r(198)`:

```
. capture noisily cd "`=subinstr("`c(filename)'", "/0_Peer_Effects_and_Rank_Concerns.do", "", .)'"
/home/datahub
...
ERROR: Stata must be started in the replication package root directory.
r(198);
```

   Commented out line 36 of `code/stata/0_Peer_Effects_and_Rank_Concerns.do`, then the code ran to completion.

4. Stata master do-file: log bug. Second run from the Stata GUI with an outer `log using "nuvolos_run.log"`; data-prep do-files 1–8 ran successfully, then `9_lca_all.do` aborted with `log file already open, r(604)` because eight subordinate do-files open their own default log via `log using ... , replace` without a preceding `capture log close`, which precludes any replicator-side outer log; closed the outer log and re-launched without one.

5. Stata master do-file: Four outputs (Figure 6 and Tables 4, A6, A12) and the second panel of Table 3 were initially missing because of 5 Windows-style separators in 15_descriptive_tables_figures.do (L582), 16_main_effects_and_rob.do (L197, L271), and 20_identifying.do (L266, L303); the replicator patched them to /, re-ran the 3 affected do-files, and confirmed all outputs were then produced.

6. Python Step 3 (`code/python/create_map.py`). Ran to completion

7. R Step 4 (`code/R/plot_histogram_earthquakes.R`). The script's own root-directory auto-detection is broken for any invocation mode other than `Rscript …` from the command line. After working around this by hard-coding the working directory, the script ran to completion.

9 Findings

The replication package **fully reproduces** the author's bundled manuscript PDF (`replication-package/Replication_Package_Tincani-20180648/Manuscript_Replication_Package_Tincani-20180648.pdf`), dated March 2026 and shipped with the replication deposit. Every figure, table, and in-text number produced by the replicator's run matches the corresponding exhibit in that PDF. See `package-output-map.xlsx` for a full list of reproduced outputs.

Assumption about the authoritative PDF. Per the replicator's correspondence with the Data Editor, the March 2026 `Manuscript_Replication_Package_Tincani-20180648.pdf` is treated as the authoritative version for this replication. The replicator also compared a selection of outputs against the earlier `paper-appendices/20251210_Peer_Effects_Tincani.pdf` (dated 10 December 2025) that was delivered with the assignment email; this older snapshot differs from the bundled PDF by a handful of observation counts. This suggests that the author re-ran the pipeline between the December 2025 submission and the March 2026 replication package.

Summary of code issues that required replicator intervention (all documented above):

1. The master do-file's `c(filename)`-based `cd` auto-detect fails on Nuvolos (redirects to `/home/datahub`, outside the package root). Disabled the auto-detect line; relied on explicit `cd` per the README.
2. The R script's `commandArgs()`-based root-directory auto-detect fails when the script is `source()`d from an R console rather than invoked via `Rscript` from the shell. Hard-coded the root for the replication run.
3. Five Windows-style `\` path separators in subordinate Stata do-files caused four output files to be missing and one appended table to be incomplete on Linux. Patched to `/` on those five lines and re-executed the three affected do-files.
4. All subordinate Stata do-files that use `log using` do so without a preceding `capture log close`. This prevents any replicator-side outer log around the master do-file, because the first subordinate `log using` then conflicts with the outer log (Stata's `r(604)`). Ran without an outer log.

9.1 Data Preparation Code

All 13 Stata data-preparation do-files (`1_…` through `13_…`) and the preparatory Python script (`calculate_coastal_proximity.py`) ran to completion and produced the expected datasets in `confidential-data-not-for-publication/processed/`.

9.2 Tables and Figures

Data Preparation Program	**Dataset created**	**Reproduced?**
code/stata/1_geog_codes_rendimiento.do	town_region_codes.dta	Yes
code/stata/2_clean_census.do	census_parents_clean_2023.dta (+ CPV2002_PERSONA/Vivienda.dta)	Yes
code/stata/3_clean_simce_4alu_8alu.do	simce_4o_2005/2007_cleaned.dta; baseline_simce_2005	Yes
code/stata/4_clean_simce_8calu.do	simce_calu_8o_2009/2011_cleaned.dta	Yes
code/stata/5_clean_4cpad_8cpad.do	baseline/endline_simce_cpad_2005/2007/2009/2011.dta	Yes
code/stata/6_clean_simce_8prof.do	endline_simce_leng/mate_2009/2011.dta	Yes
code/stata/7_clean_rendimiento.do	rendimiento_2008/2009/2010/2011_cleaned.dta	Yes
code/stata/8_merge.do	simce_cpad*_alucpad*_cleaned_2023.dta; pre/post_for_predict_2025.dta	Yes
code/stata/9_lca_all.do	census_parents_clean_wclass_all.dta (LCA); output/figures/lca_all.log (Figure 3)	Yes
code/stata/10_predict_vulnerability.do	census_with_class_type_true_predicted_all.dta; pre/post_for_predict_class_*.dta	Yes
code/python/calculate_coastal_proximity.py	towns_with_distance_to_coast.csv	Yes
code/stata/11_geog_coastal_proximity.do	comuna_alu_coastal_proximity.dta; towns_with_distance_to_coast.dta	Yes
code/stata/12_gen_variables.do	data_final_for_regs_all.dta; data_final_for_regs_teach_all.dta; prepost_cpad_alu_all.dta (main analysis dataset)	Yes
code/stata/13_gen_data_for_map.do	data_for_map_2025_all.csv; comuna_alu_geocoded.dta	Yes

Figure/Table #	Program	Line Number	Reproduced?
Figure 1	code/tex/Paper/data.tex	N/A (tikz; not generated from data)	N/A (conceptual ti
Figure 2	code/R/plot_histogram_earthquakes.R	(full script)	Yes
Figure 3	code/stata/9_lca_all.do + code/excel/donut_graph_construction_types.xlsx	9_lca_all.do line 7 (log); Excel assembled manually	Yes (manual Excel verification per §6
Figure 4	code/stata/14_analyze_census.do	201	Yes
Figure 5	code/stata/14_analyze_census.do	219	Yes
Figure 6	code/stata/15_descriptive_tables_figures.do	582	Yes (after backslash-path patch)
Figure 7	code/python/create_map.py	(full script)	Yes
Figure 8	code/stata/17_het_effects.do	628	Yes
Figure 9	code/stata/17_het_effects.do	875	Yes
Figure 10	code/stata/19_school_resources.do	479	Yes
Figure 11	code/stata/18_med_relationships.do	341	Yes

Data Preparation Program	Dataset created	Reproduced?	
Figure 12	code/tex/Paper/	N/A (conceptual drawing)	N/A (conceptual drawing)
Figure A1	code/tex/Paper/reconstruccion_dios.png	N/A (field photograph)	N/A (field photograph)
Figure A2	code/stata/17_het_effects.do	1,102	Yes
Figure A3	code/stata/18_med_relationships.do	295	Yes
Figure A4	code/stata/15_descriptive_tables_figures.do	598	Yes
Table 1	code/stata/15_descriptive_tables_figures.do	261, 293, 332 (three esttab calls appending)	Yes
Table 2	code/stata/16_main_effects_and_rob.do	54	Yes
Table 3	code/stata/20_identifying.do	213	Yes (after backslash-path patch to append)
Table 4	code/stata/20_identifying.do	303	Yes (after backslash-path patch)
Table 5	code/stata/20_identifying.do	366	Yes
Table 6	code/stata/17_het_effects.do	166	Yes
Table 7	code/stata/18_med_relationships.do	115	Yes
Table 8	code/stata/19_school_resources.do	366	Yes
Table 9	code/stata/18_med_relationships.do	215	Yes
Table A1	code/tex/Paper/appendix.tex	N/A (conceptual table)	N/A (conceptual table)
Table A2	code/stata/14_analyze_census.do	54 (outfile); 178 (write)	Yes
Table A3	code/stata/15_descriptive_tables_figures.do	482 (outfile, constructed via file write)	Yes
Table A4	code/stata/15_descriptive_tables_figures.do	549	Yes
Table A5	code/stata/16_main_effects_and_rob.do	97	Yes
Table A6	code/stata/16_main_effects_and_rob.do	197	Yes (after backslash-path patch)
Table A7	code/stata/20_identifying.do	166	Yes
Table A8	code/stata/16_main_effects_and_rob.do	150	Yes
Table A9	code/stata/17_het_effects.do	387	Yes
Table A10	code/stata/17_het_effects.do	236	Yes
Table A11	code/stata/17_het_effects.do	282	Yes
Table A12	code/stata/16_main_effects_and_rob.do	271	Yes (after backslash-path patch)
Table A13	code/stata/16_main_effects_and_rob.do	312	Yes
Table A14	code/stata/21_conley.do	76 (ts); 99 (GPA); 134 (log)	Yes (Conley not re executed; verified against author-provided Table_A14_conley

Data Preparation Program	Dataset created	Reproduced?	
			log per README §6.2)
Table A15	code/stata/16_main_effects_and_rob.do	352	Yes
Table A16	code/stata/19_school_resources.do	320	Yes
Table A17	code/stata/19_school_resources.do	259	Yes
Table A18	code/stata/19_school_resources.do	156	Yes
Table A19	code/tex/Paper/appendix_reconstruction_policy.tex	N/A (external table, translated)	N/A (external table, translated)
Table A20	code/tex/Paper/appendix_reconstruction_policy.tex	N/A (conceptual table)	N/A (conceptual table)
In-text numbers	Program	Line Number	Reproduced?
Section 2.1 (in-text)	code/stata/12_gen_variables.do	Paper §2.1 | code: 12_gen_variables.do:129 (log using)	Yes
Section 2.2 (in-text)	code/stata/14_analyze_census.do	Paper §2.2 (intro) | code: 14_analyze_census.do:21 (log using)	Yes
Sections 2.3 & 3 (in-text)	code/stata/15_descriptive_tables_figures.do	Paper §2.3 + §3 (intros) | code: 15_descriptive_tables_figures.do:49 (log using)	Yes
Section 4.1.1 (in-text)	code/stata/16_main_effects_and_rob.do	Paper §4.1.1 | code: 16_main_effects_and_rob.do:413 (log using)	Yes
Section 4.1.2 (in-text)	code/stata/18_med_relationships.do	Paper §4.1.2 | code: 18_med_relationships.do:136 (log using)	Yes
Section 4.1.2 (in-text CIs)	code/stata/18_med_relationships.do	Paper §4.1.2 (95% CIs) | code: 18_med_relationships.do:92 (file open)	Yes
Section 4.1.3 (in-text)	code/stata/19_school_resources.do	Paper §4.1.3 (intro) | code: 19_school_resources.do:228 (log using)	Yes

All 43 exhibits that are generated from data (i.e. excluding the six conceptual / external items: Figures 1, 12, A1 and Tables A1, A19, A20) were produced successfully and match the manuscript PDF. The filled `package-output-map.xlsx` reports `Yes` for every one of these, with inline notes where special treatment was required:

- **Figure 3** (donut graph): `Yes (manual Excel verification per README §6.1)`.
- **Table A14** (Conley): `Yes (Conley not re-executed; verified against author-provided` Table_A14_conley.txt `log per README §6.2)`. The replicator did not re-execute `21_conley.do`: running that do-file requires an HPC cluster with ~188 GB RAM. The log file's standard errors and significance stars for `post_st_exp`, `post_loo_mean_damage`, and `pos_loo_sd_damage` are consistent with Table A14 as printed in the manuscript.

9.3 In-Text Numbers

- [x] All in-text numbers verified.

The six in-text-numbers log files (`Section_2_1.txt`, `Section_2_2.txt`, `Sections_2_3.txt`, `Section_4_1_1.txt`, `Section_4_1_2.txt`, `Section_4_1_3.txt`) in `output/in_text_numbers/` were regenerated by the replicator's run. Every in-text claim in the corresponding paper section was located in its log file and found to match the printed value in the bundled manuscript PDF.

The `med_effects_curriculum_CIs_2025.tex` file under `output/tables/` (not a table in the paper but the source of the Section 4.1.2 in-text 95% confidence intervals) was also verified against the paper text.

10 Classification

- [x] full reproduction
- [ ] full reproduction with minor issues
- [ ] partial reproduction (see above)
- [ ] not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- [ ] None.
- [ ] `Discrepancy in output` (either figures or numbers in tables or text differ)
- [x] `Bugs in code` that were fixable by the replicator (but should be fixed in the final deposit)
- [ ] `Code missing`, in particular if it prevented the replicator from completing the reproducibility check
  - [ ] `Data preparation code missing` should be checked if the code missing seems to be data preparation code
- [ ] `Code not functional` is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- [ ] `Software not available to replicator` may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- [ ] `Insufficient time available to replicator` is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- [ ] `Data missing` is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- [ ] `Data not available` is marked when data requires additional access steps, for instance purchase or application procedure.
- [ ] `Missing README` is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements

Note: Marked `Bugs in code` because the replicator had to patch some minimal code-level issues before the pipeline ran end-to-end on Linux (see Section 13 / Replication for full detail): the master do-file's auto-detect `cd`, the R script's `commandArgs()`-based auto-detect, five Windows-style `\` path separators across three subordinate do-files.